MUGILAN SARAVANA PERUMAL

Electrical & Electronics Engineering Student | Power Electronics | EV Systems | Embedded & Hardware Design

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PROFESSIONAL SUMMARY

Power electronics and EV-focused EEE undergraduate who takes designs from concept to simulation to PCB layout. Built an EV wireless charging system, a closed-loop motor controller, and DC-DC converter hardware using MATLAB/Simulink, LTspice, KiCad, and Fusion 360. Certified by MathWorks, Autodesk, and TVS Training & Services. Seeking a Power Electronics, EV Systems, Hardware, or Embedded Systems internship.

TECHNICAL SKILLS

- **Simulation & Modeling:** MATLAB, Simulink, Simscape, LTspice, Proteus
- **Hardware & CAD:** KiCad (schematic capture, PCB layout), Autodesk Fusion 360 (3D CAD)
- **Programming:** C, C++, Python, Microsoft SQL, Front-End Web Development
- **Core Domains:** Power Electronics, Power Converters, EV Systems, Closed-Loop Control, Wireless Power Transfer

KEY PROJECTS

Smart Wireless Power Transfer System for EV Charging | *Hardware Project*

- Engineered a cable-free EV charging system, modeling the transmitter-receiver power path and magnetic coupling, and validated power-stage and control behavior in simulation across operating conditions.

Closed-Loop DC Motor Speed Control | *MATLAB, Simulink*

- Built a DC motor plant model and feedback controller for stable speed regulation; tuned the loop and verified tracking and disturbance rejection under load changes.

DC-DC Boost Converter Design | *KiCad (Schematic & PCB)*

- Took a boost converter from schematic to routed PCB, covering component selection and layout practices for switching circuits.

Buck & Boost Converter Analysis | *LTspice*

- Compared both topologies on output voltage, inductor current, ripple, and efficiency across duty cycle and load variations.

Interactive Multilingual Engineering Portfolio | *HTML, CSS, JavaScript, GitHub Pages*

- Designed and deployed a responsive multilingual website that presents my projects and skills online.

EDUCATION & ACHIEVEMENTS

B.Tech, Electrical & Electronics Engineering | 2025 - 2029 | First Class with Distinction

Sri Manakula Vinayagar Engineering College

High School Diploma | 2025 | First Class with Distinction

Amalorpavam Higher Secondary School

CERTIFICATIONS & INDUSTRY TRAINING

- **MATLAB Simulink Onramp and MATLAB Simscape Onramp**, MathWorks
- **5-Day EV Training Program**, TVS Training & Services
- **Learn Fusion for CAD in 90 Minutes**, Autodesk
- **Artificial Intelligence Traineeship**, Maiyyam